

MANROOP KAUR

Computer Science & Engineering Undergraduate | ML / AI & Data Science
Kurukshetra, Haryana | mkaur4_be24@thapar.edu | +91-9416544961 | LinkedIn | GitHub

Education

Thapar Institute of Engineering and Technology, Patiala

Aug 2024 – Present

B.E. Computer Science & Engineering — CGPA: **9.14**

DAV Public School, Kurukshetra

May 2024

Class XII (CBSE – PCME + Computer Science) — Grade: **95%**

BR International Public School, Kurukshetra

May 2022

Class X (CBSE) — Grade: **98.8%**

Technical Skills

Programming: C++, C, Python, SQL, JavaScript, HTML5, CSS3

Core CS: Object-Oriented Programming, Data Structures & Algorithms, Computer Networks, Operating Systems, DBMS, SQL

AI / ML: Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, PyTorch, OpenCV, MediaPipe, YOLO

Embedded AI & Hardware: NVIDIA Jetson Nano, Edge AI, Embedded Systems, Sensor Integration, I2C Communication, ADS1115, IoT

Software Development: React.js, Next.js, Node.js, REST APIs

Research: Data Analysis, Technical Documentation, Scientific Writing, Literature Review

Tools: VS Code, Linux, Arduino IDE, Jupyter Notebook, Git, GitHub, Canva, PowerPoint

Professional: Problem Solving, Teamwork, Leadership, Public Speaking, Project Management

Projects

DriveSafeAI 2.0 | Edge AI Driver Monitoring System

- Engineered a real-time multimodal driver monitoring system on the NVIDIA Jetson Nano, integrating computer vision with embedded sensors for intelligent driver safety monitoring.
- Developed fatigue detection using MediaPipe Face Mesh by implementing EAR, MAR, PERCLOS, blink, yawn, and head pose estimation, alongside MQ-3 alcohol sensing for multimodal driver state analysis.
- Integrated and optimized the complete embedded pipeline—including USB camera, ADS1115 ADC, buzzer, sensor interfacing, decision logic, and threshold tuning—to deliver reliable, low-latency driver alerts.

Carbon Commit | Full-Stack Carbon Emission Monitoring Platform

- Built a full-stack carbon tracking platform using React/Vite, Express, TypeScript, Prisma, and Supabase.
- Implemented SQL-based cost tracking across emission ref, activity logs, and dept master tables.

Professional Experience

Operations & Design Team Member — GeoThermX Venture

Sept 2025 – Present

- Grew GeoThermX's LinkedIn following by **356%** through **31+** posts and consistent outreach.
- Designed posters and presentation materials for two events.
- Sourced AC units and hardware components supporting prototype development.

Graphic Design Intern — Relief Mission Foundation

Nov 2025 – Jan 2026

- Designed **40+** posters and presentation materials for organizational use.
- Maintained visual consistency across digital content.

Societies & Leadership

- Secretarial Representative — Thapar MUN Society
- Volunteer — Pratigya Abhiyaan
- Member — Youth United Society
- House Captain (School)

Additional Information

Languages: English, Hindi, Punjabi

Certifications: 100 Days of Python Bootcamp – Dr. Angela Yu; McKinsey Forward Program